

Sustainable seaweed-based biopolymer films reinforced with rice husk-derived silica nanoparticles for eco-friendly packaging applications

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Highlights

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Rice husk converted into silica nanoparticles for seaweed film reinforcement.

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RH-SNPs (1–3 wt%) improved mechanical, thermal, and barrier performance of films.

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Enhanced hydrophobicity achieved with increased water contact angle at 3 wt% SNPs.

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Excessive RH-SNP loading caused agglomeration and reduced film uniformity.

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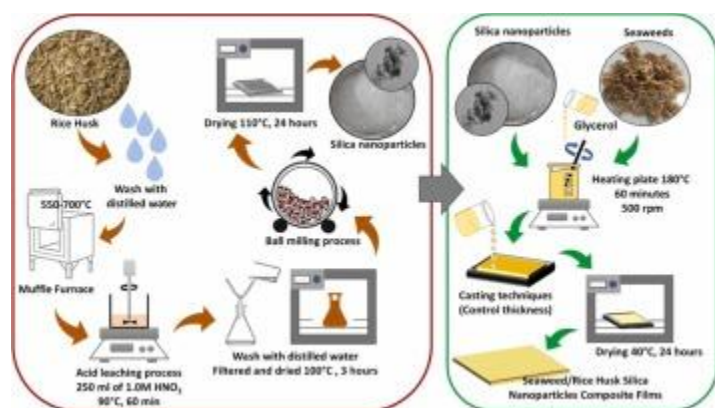
Optimized RH-SNP incorporation enables sustainable seaweed-based packaging films.

Abstract

Seaweed-derived biopolymer films are increasingly explored as sustainable alternatives to petroleum-based plastics, yet their application remains limited by low mechanical strength and poor water vapor barrier properties. In this study, silica nanoparticles synthesized from rice husk waste (RH-SNPs) were utilized as green sustainable to reinforce seaweed-based films, thereby transforming a widely available agro-residue into functional materials. The influence of RH-SNP loading (0–7 wt%) on structural, mechanical, thermal, and barrier properties was systematically investigated. Incorporation of RH-SNPs resulted in

consistent tensile strength enhancement across all nanocomposite films, with 1–3 wt% loading providing the most significant improvements due to uniform dispersion and strong interfacial hydrogen bonding with the seaweed matrix. Thermal analysis revealed increased melting temperature, enthalpy, and degradation temperature, confirming improved thermal stability at moderate filler levels. Water contact angle measurements demonstrated enhanced surface hydrophobicity, while barrier tests indicated reduced water vapor permeability due to the formation of a denser and more compact film structure. At higher loadings, nanoparticle agglomeration reduced film uniformity and performance. The 3 wt% RH-SNP loading achieved the best balance between mechanical reinforcement, thermal stability, and barrier efficiency. This work demonstrates a scalable, reproducible, and environmentally sustainable approach for producing biodegradable, high-performance films from agro-residues, offering a viable strategy for eco-friendly packaging applications.

Graphical Abstract



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Introduction

The escalating concerns over plastic pollution and finite petroleum resources have accelerated the search for the global transition toward green and biodegradable packaging materials. Conventional petroleum-based plastics such as polyethylene and polypropylene persist in the environment for decades, leading to ecological imbalance and microplastic pollution. To mitigate these impacts, the focus has shifted to biodegradable polymer composites derived from renewable resources that combine environmental compatibility with functional performance. Agro-food residues and underutilized biomass offer a sustainable pathway toward functional biopolymers, creating both value addition and waste mitigation opportunities [1]. Recent research has demonstrated that the incorporation of nanomaterials into biodegradable matrices can significantly enhance their

strength, barrier efficiency, and stability. For instance, poly(butylene adipate-co-terephthalate) (PBAT)-based films reinforced with g-C₃N₄/TiO₂ hybrids have shown improved oxidative stability and shelf-life extension in food packaging applications [2]. Similarly, PBAT composites with carbon nanoparticles and tannic acid-crosslinked carbon nanoparticle systems exhibited superior mechanical and antimicrobial performance [3], [4]. Other PBAT-based nanocomposites such as ZnO/PBAT and SnO₂/PBAT have been reported improved mechanical performance and notable antibacterial effects, illustrating how inorganic nanofillers can add multifunctionality to biodegradable matrices [5], [6].

Despite these successful demonstrations, many reported biopolymer–nanocomposite systems either rely on commercially produced nanomaterials or on polymers that are partially synthetic or derived via energy-intensive routes, which limits the overall sustainability advantage. Moreover, functional gains are frequently reported at relatively high filler contents or with additional chemical modifications that complicate scalability. For instance, active coatings combining nisin, AgNPs and montmorillonite delivered strong antimicrobial action on plasma-treated polypropylene, but such hybrid coatings address a different industrial niche (active-coated PP) rather than fully bio-derived film materials [7]. Despite these advancements, green packaging still faces major challenges in achieving a balance between mechanical performance, barrier protection, and biodegradability. Many bio-based polymers such as PBAT, PLA, and PHA, while compostable, depend on costly synthesis routes or limited natural feedstocks.

In this context, agro-food residues and underutilized biomass present abundant, low-cost, and sustainable alternatives [8]. Among these, seaweed has recently gained prominence as a promising feedstock for biopolymer film production due to its rapid cultivation, minimal resource requirements, and non-competition with land-based crops. Unlike terrestrial starch or protein sources, seaweed cultivation requires no arable land, freshwater, or fertilizers, making it highly compatible [9]. Its polysaccharide-rich structure and marine origin make it especially appealing, as it does not compete with land crops and requires no freshwater or fertilizers for cultivation. Compounds extracted from seaweed, including carrageenan, alginate, and agar possess excellent film-forming abilities and have been extensively investigated for their mechanical strength, thermal stability, and barrier performance [10], [11], [12].

Seaweed species are classified based on their pigment profiles into brown (Heterokontophyta), red (Rhodophyta), and green (Chlorophyta) algae [13]. Each group contains unique chlorophyll and accessory pigments, contributing to their functionality and extractable polysaccharide content [14], [15]. In particular, Malaysia, especially the state of Sabah, is one of the world's top producers of red seaweed used for carrageenan

production. While most prior research focuses on extracted derivatives, the direct use of raw seaweed biomass for film formation remains limited, despite offering simplified processing and cost reduction by avoiding extraction and purification steps. However, raw seaweed-based films exhibit inherently low mechanical strength, thermal resistance, and water vapor barrier properties compared to synthetic polymers [16]. To address these challenges, reinforcement with nanoscale fillers has emerged as a viable strategy for enhancing performance [17], [18].

Among the various nanofillers, silica nanoparticles (SNPs) are particularly attractive owing to their abundance, non-toxicity, high surface area, and ability to strengthen both mechanical and barrier properties of biopolymer films [19], [20]. While conventional synthesis routes exist, deriving silica from agricultural residues provides a more sustainable and low-cost alternative. Rice husk (RH), a major byproduct of paddy cultivation, is generated in millions of tons annually in Southeast Asia but is underutilized due to its low bulk density and disposal challenges [21], [22], [23]. For instance, Malaysia produces approximately 0.77 million tons of rice husk per year, while neighbouring countries such as Thailand, Vietnam, and Indonesia collectively generate over 60 million tons annually, and around 104 million tons globally, highlighting the regional abundance of this agro-residue [22], [24], [25]. Controlled combustion of RH yields ash rich in amorphous silica (80–95 %), associated with higher purity and larger surface areas positioning it as an abundant and renewable precursor for nanoparticle synthesis [21], [26]. The methodology is reproducible across different RH sources and is extendable to other biomass-derived residues, ensuring that the approach is not limited to a local region. Valorizing rice husk into silica nanoparticles not only addresses disposal issues but also supports circular economy practices within agricultural supply chains.

Studies have demonstrated that incorporation of SNPs into biopolymers such as starch or chitosan improves tensile strength, thermal and water resistance [27], [28]. The enhancement in mechanical performance is attributed to the robust interaction at the interface between the nanoparticles and the polymer matrix, which facilitates efficient stress distribution and improves overall structural stability. Recent work has further demonstrated the efficacy of seaweed-based nanocomposite films for green packaging, such as carrageenan-based films with embedded nanomaterials showing significant improvements in mechanical strength (>16 MPa), thermal stability (>180 °C), and barrier performance while maintaining full biodegradability within a short timeframe [29]. However, the integration of RH-derived silica nanoparticles (RH-SNPs) into raw seaweed-based films has not been widely reported.

Therefore, the present study aims to fabricate novel seaweed-based biopolymer films reinforced with RH-SNPs synthesized in the laboratory from agricultural waste. Unlike previous studies that relied on commercial nanofillers or single-source biopolymers, this work introduces a dual-waste valorization approach by coupling marine biomass (seaweed) with valorized agro-waste (rice husk). The resulting nanocomposite films were systematically characterized for their physical, mechanical, thermal, physicochemical, morphological, and barrier properties. This integrated design not only demonstrates the feasibility of transforming underutilized waste resources into functional materials but also provides a sustainable pathway for producing high-performance, eco-friendly packaging films with enhanced strength, thermal stability, and water vapor resistance.